

EEE 554 Homework 1

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Problem 2-6 (Page 44)

Problem Statement: Show that if S consists of a countable number of elements ζ_i and each subset $\{\zeta_i\}$ is an event, then all subsets of S are events.

Proof. Let $S = \{\zeta_1, \zeta_2, \dots\}$ be a countable sample space. By definition, the collection of all events forms a **Borel field**. A defining property of a Borel field $\mathcal{F}$ is that it is closed under countable unions. That is, if $A_1, A_2, \dots \in \mathcal{F}$, then:

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$$

Let A be an arbitrary subset of S . Since S is countable, any subset A consists of a countable collection of elements from S . Let K be the index set of the elements contained in A . We can express A as the union of its constituent singleton sets:

$$A = \bigcup_{i \in K} \{\zeta_i\}$$

We are given that every singleton set $\{\zeta_i\}$ is an event, which means $\{\zeta_i\} \in \mathcal{F}$ for all i .

Since A is formed by a countable union of events $\{\zeta_i\}$, and the field of events $\mathcal{F}$ is closed under countable union, it follows that A is also an event ($A \in \mathcal{F}$).

Therefore, all subsets of S are events. $\square$

Problem 2-7 (Page 44)

Problem Statement: If $S = \{1, 2, 3, 4\}$, find the smallest field that contains the sets $\{1\}$ and $\{2, 3\}$.

Solution

Let the given sets be $A = \{1\}$ and $B = \{2, 3\}$. To construct the smallest field $\mathcal{F}$, we must ensure it is closed under complements and unions. By the properties of a field, we can derive the following sets:

1. We have $\{1\} \in \mathcal{F}$ and $\{2, 3\} \in \mathcal{F}$.
2. The union of these sets is $\{1, 2, 3\} \in \mathcal{F}$.
3. The complement of this union with respect to $S = \{1, 2, 3, 4\}$ must also be in $\mathcal{F}$:

$$\overline{\{1, 2, 3\}} = \{4\}$$

Thus, the partition of S is:

$$E_1 = \{1\}, \quad E_2 = \{2, 3\}, \quad E_3 = \{4\}$$

The smallest field consists of all possible unions of these three sets. There are $2^3 = 8$ such combinations:

$$\mathcal{F} = \left\{ \begin{array}{ll} \emptyset, & (\text{Empty set}) \\ \{1\}, & (E_1) \\ \{2, 3\}, & (E_2) \\ \{4\}, & (E_3) \\ \{1, 2, 3\}, & (\text{Union } E_1 \cup E_2) \\ \{1, 4\}, & (\text{Union } E_1 \cup E_3) \\ \{2, 3, 4\}, & (\text{Union } E_2 \cup E_3) \\ \{1, 2, 3, 4\} & (\text{Whole space } S) \end{array} \right\}$$

Problem 2-10 (Page 45)

Problem Statement: (Chain rule) Show that

$$P(A_n \dots A_1) = P(A_n | A_{n-1} \dots A_1) \dots P(A_2 | A_1) P(A_1)$$

Proof. We use the definition of conditional probability, $P(AB) = P(A|B)P(B)$, iteratively.

Consider the joint probability of n events, denoted as the intersection $P(A_n \cap A_{n-1} \cap \dots \cap A_1)$. Let $B = A_{n-1} \dots A_1$. Then:

$$P(A_n A_{n-1} \dots A_1) = P(A_n | A_{n-1} \dots A_1) P(A_{n-1} \dots A_1)$$

Next, we expand the term $P(A_{n-1} \dots A_1)$ by conditioning A_{n-1} on the remaining events $A_{n-2} \dots A_1$:

$$P(A_{n-1} \dots A_1) = P(A_{n-1} | A_{n-2} \dots A_1) P(A_{n-2} \dots A_1)$$

Substituting this back into the first equation:

$$P(A_n \dots A_1) = P(A_n | A_{n-1} \dots A_1) P(A_{n-1} | A_{n-2} \dots A_1) P(A_{n-2} \dots A_1)$$

Repeating this process $n - 1$ times, we can reach the last term $P(A_1)$:

$$\begin{aligned} P(A_n \dots A_1) &= P(A_n | A_{n-1} \dots A_1) \\ &\quad \times P(A_{n-1} | A_{n-2} \dots A_1) \\ &\quad \times \dots \\ &\quad \times P(A_2 | A_1) \\ &\quad \times P(A_1) \end{aligned}$$

This completes the proof of the chain rule. □

Problem 4

Statement: Consider events A and B that are exhaustive in S with $P(B) = \alpha P(A)$ where $\alpha \geq 1$. If A and B are independent, show that $P(B) = 1$.

Proof. We are given three conditions:

1. A and B are exhaustive: $P(A \cup B) = 1$.
2. Independence: $P(AB) = P(A)P(B)$.
3. Ratio: $P(B) = \alpha P(A)$ with $\alpha \geq 1$.

Using the addition law of probability:

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

Substituting the given values:

$$1 = P(A) + P(B) - P(A)P(B)$$

Rearranging terms to factor:

$$1 - P(B) = P(A) - P(A)P(B)$$

$$1 - P(B) = P(A)[1 - P(B)]$$

$$[P(A) - 1][1 - P(B)] = 0$$

This equation implies two possible cases:

- **Case 1:** $P(B) = 1$. In this case, the proof is complete.
- **Case 2:** $P(A) = 1$. If $P(A) = 1$, then $P(B) = \alpha P(A) = \alpha$. Since $P(B)$ is a probability, $P(B) \leq 1$. Given $\alpha \geq 1$, the only valid solution is $\alpha = 1$, which implies $P(B) = 1$.

Thus, in all valid cases, $P(B) = 1$. □

Problem 2-13 (Page 45)

Statement: The space S is the set of all positive numbers t . Show that if

$$P\{t_0 \leq t \leq t_0 + t_1 | t \geq t_0\} = P\{t \leq t_1\}$$

for every t_0 and t_1 , then $P\{t \leq t_1\} = 1 - e^{-ct_1}$.

Proof. Let T be the random variable representing the outcome (e.g., time to failure). Let $G(t)$ be the **Reliability Function** (or Survival Function), defined as the probability that the variable exceeds t :

$$G(t) = P\{T > t\}$$

Consequently, the Cumulative Distribution Function (CDF) is:

$$F(t) = P\{T \leq t\} = 1 - G(t)$$

The problem statement gives the condition regarding the probability of failure in an interval $(t_0, t_0 + t_1)$, given survival up to t_0 . It is mathematically cleaner to work with the complementary event (survival).

1. Translate the Condition to Survival Probability The given condition is:

$$P\{t_0 \leq T \leq t_0 + t_1 \mid T \geq t_0\} = P\{T \leq t_1\}$$

The LHS represents the probability of “dying” in the interval $[t_0, t_0 + t_1]$. The complementary event is “surviving” past $t_0 + t_1$, given $T \geq t_0$.

$$P\{T > t_0 + t_1 \mid T \geq t_0\} = 1 - P\{t_0 \leq T \leq t_0 + t_1 \mid T \geq t_0\}$$

Substituting the given condition into the RHS:

$$P\{T > t_0 + t_1 \mid T \geq t_0\} = 1 - P\{T \leq t_1\} = P\{T > t_1\}$$

Thus, the problem condition is equivalent to the memoryless property:

$$P\{T > t_0 + t_1 \mid T \geq t_0\} = P\{T > t_1\} \quad (1)$$

2. Apply Conditional Probability Definition Using the definition $P(A|B) = \frac{P(AB)}{P(B)}$:

$$\frac{P\{T > t_0 + t_1 \cap T \geq t_0\}}{P\{T \geq t_0\}} = P\{T > t_1\}$$

Since the event $\{T > t_0 + t_1\}$ is a subset of $\{T \geq t_0\}$, their intersection is simply $\{T > t_0 + t_1\}$. Also, using our definition $G(t) = P\{T > t\}$:

$$\frac{G(t_0 + t_1)}{G(t_0)} = G(t_1)$$

3. Solve the Functional Equation Rearranging the terms gives:

$$G(t_0 + t_1) = G(t_0)G(t_1)$$

This is the Cauchy Functional Equation. For a bounded, monotonic function (since probability must be between 0 and 1), the only non-trivial continuous solution is the exponential function:

$$G(t) = e^{-ct}$$

where c is a positive constant ($c > 0$).

4. Final Result Substituting back into the definition of the CDF:

$$P\{T \leq t\} = 1 - G(t) = 1 - e^{-ct}$$

□

Problem 4-8 (Page 120)

Statement: The random variable x is $N(10, 1)$. Find $f(x|(x - 10)^2 < 4)$.

Solution

Let M be the event defined by the condition $(x - 10)^2 < 4$. This inequality is equivalent to:

$$|x - 10| < 2 \implies -2 < x - 10 < 2 \implies 8 < x < 12$$

The conditional PDF is defined as the truncated original PDF, normalized by the probability of the conditioning event:

$$f(x|M) = \begin{cases} \frac{f_x(x)}{P(M)} & 8 < x < 12 \\ 0 & \text{otherwise} \end{cases}$$

First, we calculate $P(M)$. Since $x \sim N(10, 1)$, the variable $z = x - 10$ is standard normal $N(0, 1)$.

$$P(M) = P(8 < x < 12) = P(-2 < z < 2)$$

Using the symmetry of the standard normal distribution $\Phi(z)$:

$$P(-2 < z < 2) = \Phi(2) - \Phi(-2) = \Phi(2) - (1 - \Phi(2)) = 2\Phi(2) - 1$$

Using standard tables, $\Phi(2) \approx 0.9772$.

$$P(M) \approx 2(0.9772) - 1 = 0.9544$$

The original PDF for $N(10, 1)$ is:

$$f_x(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-10)^2}{2}}$$

Thus, the conditional density is:

$$f(x|(x - 10)^2 < 4) = \frac{1}{(2\Phi(2) - 1)\sqrt{2\pi}} e^{-\frac{(x-10)^2}{2}}, \quad 8 < x < 12$$

Substituting the numerical value for the normalization constant:

$$f(x|M) \approx \frac{1}{0.9544\sqrt{2\pi}} e^{-\frac{(x-10)^2}{2}} \approx 0.418 e^{-\frac{(x-10)^2}{2}}, \quad 8 < x < 12$$

and 0 otherwise.

Problem 7

Statement: Define $h(x)$ as:

$$h(x) = \begin{cases} c & \text{if } x \in (n, n + 2^{-(n+1)}) \text{ for } n = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

(a) Find a value of c for which h is a PDF.

For $h(x)$ to be a PDF, the integral over the entire domain must be 1.

$$\int_{-\infty}^{\infty} h(x) dx = \sum_{n=0}^{\infty} \text{Area}_n = 1$$

The area of the n -th interval is given by height $\times$ width:

$$\text{Area}_n = c \cdot 2^{-(n+1)}$$

Summing these areas (a geometric series):

$$\sum_{n=0}^{\infty} c \left(\frac{1}{2}\right)^{n+1} = c \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k = c \left(\frac{1/2}{1 - 1/2}\right) = c(1) = c$$

Setting the total area to 1, we find:

$$c = 1$$

(b) Calculate $E[x]$.

The expected value is:

$$E[x] = \int_{-\infty}^{\infty} xh(x) dx = \sum_{n=0}^{\infty} \int_n^{n+2^{-(n+1)}} x \cdot (1) dx$$

Evaluating the integral for a single interval:

$$\int_a^{a+\delta} x dx = \frac{(a+\delta)^2 - a^2}{2} = a\delta + \frac{\delta^2}{2}$$

Letting $a = n$ and $\delta = 2^{-(n+1)}$:

$$\begin{aligned} E[x] &= \sum_{n=0}^{\infty} \left[n \left(\frac{1}{2}\right)^{n+1} + \frac{1}{2} \left(\frac{1}{2}\right)^{2n+2} \right] \\ &= \frac{1}{2} \sum_{n=0}^{\infty} n \left(\frac{1}{2}\right)^n + \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^{n+1} \end{aligned}$$

Using the series identities $\sum nx^n = x/(1-x)^2$ and $\sum x^{n+1} = x/(1-x)$:

- First term: $\frac{1}{2} \cdot \frac{1/2}{(1/2)^2} = \frac{1}{2} \cdot 2 = 1$
- Second term: $\frac{1}{2} \cdot \frac{1/4}{3/4} = \frac{1}{6}$

$$E[x] = 1 + \frac{1}{6} = \frac{7}{6}$$

(c) Construct a PDF with arbitrarily large values on tails.

We construct a “comb” of rectangular pulses that get taller and thinner as y increases. Let the PDF be non-zero on intervals centered at integers $n > 0$. Define the height of the pulse at n to be $h_n = n$ and the width to be $w_n = \frac{1}{n^3}$. The area of the n -th pulse is $A_n = n \cdot \frac{1}{n^3} = \frac{1}{n^2}$. Since $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty$, we can normalize this sum to create a valid PDF.

Let $C = \frac{6}{\pi^2}$. Define:

$$f_y(y) = \begin{cases} C \cdot n & \text{if } y \in [n, n + n^{-3}] \text{ for } n \in \mathbb{Z}^+ \\ 0 & \text{otherwise} \end{cases}$$

For any b , there exists an integer $n > b$. At this location, $f_y(y) = C \cdot n$. As $n \rightarrow \infty$, $f_y(y) \rightarrow \infty$. Thus, the function takes on arbitrarily large values on the tails while maintaining a finite integral of 1.

Problem 5-1 (Page 164)

Problem Statement: The random variable x is $N(5, 2)$ and $y = 2x + 4$. Find η_y , σ_y^2 , and $f_y(y)$.

Solution

Given:

- $x \sim N(\eta_x, \sigma_x)$ with $\eta_x = 5$ and $\sigma_x = 2$.
- Transformation: $y = 2x + 4$.

1. Mean of y (η_y) Using the linearity of expectation $E[ax + b] = aE[x] + b$:

$$\eta_y = E[2x + 4] = 2E[x] + 4 = 2(5) + 4 = 14$$

2. Variance of y (σ_y^2) Using the property of variance $\text{Var}(ax + b) = a^2 \text{Var}(x)$: The variance of x is $\sigma_x^2 = 2^2 = 4$.

$$\sigma_y^2 = 2^2 \sigma_x^2 = 4(4) = 16$$

(Standard deviation $\sigma_y = \sqrt{16} = 4$).

3. PDF of y ($f_y(y)$) Since x is a Gaussian random variable, the linear transformation $y = 2x+4$ is also a Gaussian random variable with mean $\eta_y = 14$ and variance $\sigma_y^2 = 16$. The general form of the Gaussian PDF is:

$$f_y(y) = \frac{1}{\sqrt{2\pi}\sigma_y} e^{-\frac{(y-\eta_y)^2}{2\sigma_y^2}}$$

Substituting the calculated values:

$$f_y(y) = \frac{1}{\sqrt{2\pi}(4)} e^{-\frac{(y-14)^2}{2(16)}}$$

$$f_y(y) = \frac{1}{4\sqrt{2\pi}} e^{-\frac{(y-14)^2}{32}}$$

Problem 9

Statement: The random variable x has PDF

$$f_x(x) = \begin{cases} c(a^2 - x^2) & |x| < a \\ 0 & \text{otherwise} \end{cases}$$

(a) Find c in terms of a .

The total probability must integrate to 1:

$$\int_{-a}^a c(a^2 - x^2) dx = 1$$

Using symmetry (even function):

$$2c \int_0^a (a^2 - x^2) dx = 2c \left[a^2x - \frac{x^3}{3} \right]_0^a = 2c \left(a^3 - \frac{a^3}{3} \right) = \frac{4ca^3}{3}$$

Setting this equal to 1:

$$\frac{4ca^3}{3} = 1 \implies c = \frac{3}{4a^3}$$

(b) Calculate $P(\{-a < 4x < 3a\})$.

The inequality simplifies to $-\frac{a}{4} < x < \frac{3a}{4}$.

$$P = \int_{-a/4}^{3a/4} \frac{3}{4a^3} (a^2 - x^2) dx = \frac{3}{4a^3} \left[a^2 x - \frac{x^3}{3} \right]_{-a/4}^{3a/4}$$

Evaluation at limits:

- Upper $(3a/4)$: $a^2(\frac{3a}{4}) - \frac{1}{3}(\frac{3a}{4})^3 = \frac{3a^3}{4} - \frac{9a^3}{64} = \frac{39a^3}{64}$
- Lower $(-a/4)$: $a^2(-\frac{a}{4}) - \frac{1}{3}(-\frac{a}{4})^3 = -\frac{a^3}{4} + \frac{a^3}{192} = -\frac{47a^3}{192}$

$$P = \frac{3}{4a^3} \left(\frac{39a^3}{64} - \left(-\frac{47a^3}{192} \right) \right) = \frac{3}{4} \left(\frac{117 + 47}{192} \right) = \frac{3}{4} \cdot \frac{164}{192}$$

$$P = \frac{3}{4} \cdot \frac{41}{48} = \frac{41}{64}$$

(c) Find mean and variance of x .

Mean: Since $f_x(x)$ is an even function on a symmetric interval $[-a, a]$, the expected value $E[x]$ involves integrating an odd function $xf_x(x)$, which is zero.

$$E[x] = 0$$

Variance: $Var(x) = E[x^2] - (E[x])^2 = E[x^2]$.

$$E[x^2] = \int_{-a}^a x^2 \cdot c(a^2 - x^2) dx = 2c \int_0^a (a^2 x^2 - x^4) dx$$

$$= 2c \left[\frac{a^2 x^3}{3} - \frac{x^5}{5} \right]_0^a = 2c \left(\frac{a^5}{3} - \frac{a^5}{5} \right) = 2c \left(\frac{2a^5}{15} \right) = \frac{4ca^5}{15}$$

Substitute $c = \frac{3}{4a^3}$:

$$Var(x) = \frac{4a^5}{15} \cdot \frac{3}{4a^3} = \frac{12a^5}{60a^3} = \frac{1}{5}a^2$$

Problem 10

Problem Statement: A random variable x is Gamma distributed. Calculate the moments $E[x^k]$ for $k = 1, 2, \dots$ using the characteristic function.

Solution

1. Characteristic Function of Gamma Distribution The PDF of a Gamma random variable with parameters α and λ is:

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} u(x)$$

The characteristic function $\Phi_x(\omega)$ is defined as $E[e^{j\omega x}]$:

$$\Phi_x(\omega) = \int_0^\infty e^{j\omega x} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\lambda-j\omega)x} dx$$

Using the definition of the Gamma integral, this evaluates to:

$$\Phi_x(\omega) = \frac{\lambda^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha)}{(\lambda-j\omega)^\alpha} = \left(1 - \frac{j\omega}{\lambda}\right)^{-\alpha}$$

2. Moment Generation Formula The k -th moment is given by the derivative property of the characteristic function:

$$E[x^k] = \frac{1}{j^k} \frac{d^k \Phi_x(\omega)}{d\omega^k} \Big|_{\omega=0}$$

3. Derivation Differentiating $\Phi_x(\omega) = (1 - j\omega/\lambda)^{-\alpha}$ repeatedly:

• **First Derivative:**

$$\frac{d\Phi}{d\omega} = -\alpha \left(1 - \frac{j\omega}{\lambda}\right)^{-\alpha-1} \left(-\frac{j}{\lambda}\right) = \frac{j\alpha}{\lambda} \left(1 - \frac{j\omega}{\lambda}\right)^{-\alpha-1}$$

• **Second Derivative:**

$$\frac{d^2\Phi}{d\omega^2} = \frac{j\alpha}{\lambda} (-\alpha-1) \left(1 - \frac{j\omega}{\lambda}\right)^{-\alpha-2} \left(-\frac{j}{\lambda}\right) = \left(\frac{j}{\lambda}\right)^2 \alpha(\alpha+1) \left(1 - \frac{j\omega}{\lambda}\right)^{-\alpha-2}$$

• **k -th Derivative:** Following the pattern, the k -th derivative is:

$$\frac{d^k\Phi}{d\omega^k} = \left(\frac{j}{\lambda}\right)^k \alpha(\alpha+1) \cdots (\alpha+k-1) \left(1 - \frac{j\omega}{\lambda}\right)^{-\alpha-k}$$

4. Evaluation at $\omega = 0$

$$\frac{d^k\Phi}{d\omega^k} \Big|_{\omega=0} = j^k \frac{\alpha(\alpha+1) \cdots (\alpha+k-1)}{\lambda^k}$$

Substituting this back into the moment formula:

$$E[x^k] = \frac{1}{j^k} \left[j^k \frac{\alpha(\alpha+1) \cdots (\alpha+k-1)}{\lambda^k} \right]$$

$$E[x^k] = \frac{\alpha(\alpha+1)\cdots(\alpha+k-1)}{\lambda^k}$$

Using the property of the Gamma function where $\Gamma(z+1) = z\Gamma(z)$, we can write the numerator as $\frac{\Gamma(\alpha+k)}{\Gamma(\alpha)}$.

Final Result:

$$E[x^k] = \frac{\Gamma(\alpha+k)}{\lambda^k \Gamma(\alpha)}$$

Problem 11

Problem Statement: (Computer problem) Use a random number generator to generate 10,000 data values drawn independently from a normal distribution with mean zero and variance four. Similarly, obtain 10,000 data drawn independently from a uniform distribution with mean zero and variance four. (a) Produce histograms of both data sets. (b) For each of the two distributions, calculate the probability that a datum falls between 1 and 2. Compare these theoretical figures to the actual percentages of data that fall in this range in each data set. (c) Perform a two-sample Kolmogorov-Smirnov goodness of fit test on the two data sets and interpret the result.

Solution

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy import stats
4
5 # Set a random seed for reproducibility
6 np.random.seed(42)
7
8 # --- 1. Generate data ---
9 N = 10000
10
11 # (a) Normal distribution
12 # Mean = 0, variance = 4 => standard deviation (scale) = sqrt(4) =
13 # 2
14 data_norm = np.random.normal(loc=0, scale=2, size=N)
15
16 # (b) Uniform distribution
17 # Mean = 0 => interval is [-b, b]
18 # Variance = 4 => (2b)^2 / 12 = 4 => 4b^2 = 48 => b = sqrt(12)
19 # approx 3.464
20 limit = np.sqrt(12)
21 data_unif = np.random.uniform(low=-limit, high=limit, size=N)
22
23 # --- 2. (a) Plot histograms ---

```

```

22 plt.figure(figsize=(12, 5))
23
24 # Histogram for the normal distribution
25 plt.subplot(1, 2, 1)
26 plt.hist(
27     data_norm, bins=50, density=True, alpha=0.7, color="skyblue",
28     edgecolor="black"
29 )
30 plt.title("Normal Distribution (Mean=0, Var=4)")
31 plt.xlabel("Value")
32 plt.ylabel("Density")
33
34 # Histogram for the uniform distribution
35 plt.subplot(1, 2, 2)
36 plt.hist(
37     data_unif, bins=50, density=True, alpha=0.7, color="lightgreen",
38     edgecolor="black"
39 )
40 plt.title("Uniform Distribution (Mean=0, Var=4)")
41 plt.xlabel("Value")
42 plt.ylabel("Density")
43
44 plt.tight_layout()
45 plt.show()
46
47 # --- 3. (b) Compute the probability of falling in the interval [1,
48 # 2] ---
49 # Theoretical value for the normal distribution
50 prob_norm_theo = stats.norm.cdf(2, loc=0, scale=2) - stats.norm.cdf(
51     1, loc=0, scale=2)
52 # Empirical value for the normal distribution
53 prob_norm_emp = np.sum((data_norm >= 1) & (data_norm <= 2)) / N
54
55 # Theoretical value for the uniform distribution
56 # Interval length is 1, total support length is 2*sqrt(12)
57 prob_unif_theo = (2 - 1) / (2 * limit)
58 # Empirical value for the uniform distribution
59 prob_unif_emp = np.sum((data_unif >= 1) & (data_unif <= 2)) / N
60
61 print(
62     f"Normal Distribution [1, 2]: Theoretical = {prob_norm_theo:.4f}
63     }, Empirical = {prob_norm_emp:.4f}"
64 )
65 print(
66     f"Uniform Distribution [1, 2]: Theoretical = {prob_unif_theo:.4f}
67     }, Empirical = {prob_unif_emp:.4f}"
68 )
69
70 # --- 4. (c) K-S test (Kolmogorov-Smirnov Test) ---
71 # Test whether the two samples come from the same distribution
72 ks_stat, p_value = stats.ks_2samp(data_norm, data_unif)
73
74 print(f"\nK-S Test Result:")
75 print(f"Statistic = {ks_stat:.4f}")
76 print(f"P-value = {p_value:.4e}")

```

Listing 1: Python Simulation for Problem 11

1. Theoretical Parameters and Probabilities First, we establish the parameters for both distributions given a mean $\mu = 0$ and variance $\sigma^2 = 4$.

- **Normal Distribution $N(0, 4)$:** The standard deviation is $\sigma = \sqrt{4} = 2$. The theoretical probability of falling in the interval $[1, 2]$ is:

$$P_{norm} = P(1 \leq X \leq 2) = \Phi\left(\frac{2-0}{2}\right) - \Phi\left(\frac{1-0}{2}\right) = \Phi(1) - \Phi(0.5)$$

Using standard normal tables ($\Phi(1) \approx 0.8413$, $\Phi(0.5) \approx 0.6915$):

$$P_{norm} \approx 0.8413 - 0.6915 = 0.1498$$

- **Uniform Distribution $U[a, b]$:** Since the mean is 0, the interval is symmetric: $[-L, L]$. The variance is $\frac{(2L)^2}{12} = 4 \implies \frac{4L^2}{12} = 4 \implies L^2 = 12 \implies L = \sqrt{12} \approx 3.464$. The PDF is $f(x) = \frac{1}{2\sqrt{12}}$ for $x \in [-\sqrt{12}, \sqrt{12}]$. The theoretical probability of falling in $[1, 2]$ is:

$$P_{unif} = \int_1^2 \frac{1}{2\sqrt{12}} dx = \frac{2-1}{2\sqrt{12}} = \frac{1}{4\sqrt{3}} \approx 0.1443$$

2. Simulation Logic (Python/Matlab) A simulation was performed generating $N = 10,000$ points for each distribution.

- **Data Generation:** Normal data generated with $\mu = 0, \sigma = 2$. Uniform data generated in range $[-\sqrt{12}, \sqrt{12}]$.
- **Empirical Probability Calculation:** Counted number of samples x where $1 \leq x \leq 2$ and divided by N .
- **Kolmogorov-Smirnov Test:** Computed the statistic D and p -value to test the null hypothesis that both samples come from the same distribution.

3. Simulation Results

- **(a) Histograms:** The Normal data produced a bell-shaped histogram concentrated near 0. The Uniform data produced a flat, rectangular histogram extending from ≈ -3.46 to 3.46 .
- **(b) Probability Comparison:** The empirical results from the simulation closely matched the theoretical derivations:

$$\text{Normal: Theoretical} = 0.1499, \quad \text{Empirical} \approx 0.1470$$

$$\text{Uniform: Theoretical} = 0.1443, \quad \text{Empirical} \approx 0.1487$$

(Note: Empirical values vary slightly per simulation run.)

- (c) **K-S Test Result:** The two-sample K-S test yielded:

$$\text{Statistic } D \approx 0.0705$$

$$\text{P-value } p \approx 5.0 \times 10^{-22}$$

4. Interpretation The K-S test compares the cumulative distribution functions (CDFs) of the two datasets.

$$p\text{-value} \ll 0.05$$

Since the p -value is effectively zero, we **reject the null hypothesis**. **Conclusion:** Although both distributions share the same mean (0) and variance (4) and have very similar probabilities for the specific interval $[1, 2]$ (both $\approx 14.5\%$), the test confirms they are statistically significantly different distributions.